Intelligent Control

Session 6: Energy, Lyapunov Functions, and Lyapunov-Based Design

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Today

1. The energy of the pendulum, and how it changes — *the first Lyapunov function*
2. Lyapunov's idea: a bowl that the motion can only go down
3. LaSalle's argument: when $\dot{V} \leq 0$ is not quite enough
4. Designing with V: hold any angle; swing up with a weak motor
5. Simulation lab

Learning objectives

- compute $\dot{V}$ along the motion of a system, using the chain rule;
- decide stability with a Lyapunov function, and know what it does *not* tell you;
- use LaSalle's argument to show that the error goes to zero when $\dot{V} \leq 0$;
- design a controller by choosing V first and then u so that $\dot{V} \leq 0$.

Before we start: get the code

Every example and exercise of today runs from **one** Python file:

`pendulum_lyapunov.py`

1. Go to the course repository on GitHub: `github.com/gfloresc/intelligent-control`
2. Open the folder `code/` and download `pendulum_lyapunov.py`.
3. Open it in Google Colab and run it. You should see the pendulum swing back and forth, higher each time, until it stops upright, with a 1.2 N m motor.

Same structure as Sessions 4 and 5: a list `SCENARIOS` near the top, one entry per example of the slides, and a single line to change,

`SCENARIO = 'swing_up'` $\longrightarrow$ `SCENARIO = 'free'`.

The animation shows the pendulum and plots three curves against time: the angle θ, the function V of the slides (energy or Lyapunov function), and the torque u.

In `pendulum_lyapunov.py`, mark `[ENERGY] ... [CATCH]`: the grey boxes on the slides tell you which scenario to run and which lines to look at. Search the file for the mark.

Where we left off

Session 5 ended with two open questions.

1. Proving that the error goes to zero, without simulating. With a perfect model the error obeyed $\ddot{e} + k_d \dot{e} + k_p e = 0$, and we proposed an "energy of the error"

$$V = \tfrac{1}{2}\dot{e}^2 + \tfrac{1}{2}k_p e^2, \qquad \dot{V} = -k_d\,\dot{e}^2 \leq 0.$$

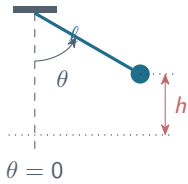
But $\dot{V} = 0$ whenever $\dot{e} = 0$, even if $e \neq 0$. Is that enough to say $e \to 0$?

2. The weak motor. Following a swing-up path needs 1.55 N m at the horizontal. With a 1.2 N m motor, computed torque saturated and the pendulum never reached the top: that path, with that controller, is out of reach for this motor. Is the pendulum then impossible to swing up with 1.2 N m?

Today's answer to both

Think in terms of **energy**. A function that can only go down, plus one more argument (LaSalle), tells us where the system ends without solving the equations (question 1). And a controller that pumps energy, instead of forcing a path, swings the pendulum up with 1.2 N m, even with 0.6 N m (question 2).

The energy of the pendulum



$h = \ell(1 - \cos\theta)$: height above the lowest point.

$$E(\theta, \dot\theta) = \underbrace{\tfrac{1}{2}\, m\ell^2\dot\theta^2}_{\text{kinetic}} + \underbrace{mg\ell\,(1 - \cos\theta)}_{\text{potential, } mgh}$$

- hanging at rest ($\theta = 0,\ \dot\theta = 0$): $E = 0$;
- upright at rest ($\theta = \pi,\ \dot\theta = 0$): $E_{top} = 2mg\ell = 2.943\,\text{J}$;
- $E \geq 0$ always, and $E = 0$ only hanging at rest ($\theta = 0 \bmod 2\pi$).

Same pendulum as always: $m = 0.5\,\text{kg}$, $\ell = 0.3\,\text{m}$, $b = 0.05$, so $m\ell^2 = 0.045$ and $mg\ell = 1.4715$.

Question. The motor and friction act on the pendulum. How does E change in time?

How the energy changes: the chain rule

E depends on time only through $\theta(t)$ and $\dot\theta(t)$. The chain rule gives

Step 1. $\dot E = \dfrac{\partial E}{\partial \theta}\,\dot\theta + \dfrac{\partial E}{\partial \dot\theta}\,\ddot\theta = mg\ell\sin\theta\,\dot\theta + m\ell^2\dot\theta\,\ddot\theta.$

Step 2. Replace $\ddot\theta$ by the model, $m\ell^2\ddot\theta = -mg\ell\sin\theta - b\dot\theta + u$:

$$\dot E = mg\ell\sin\theta\,\dot\theta + \dot\theta\big(-mg\ell\sin\theta - b\dot\theta + u\big).$$

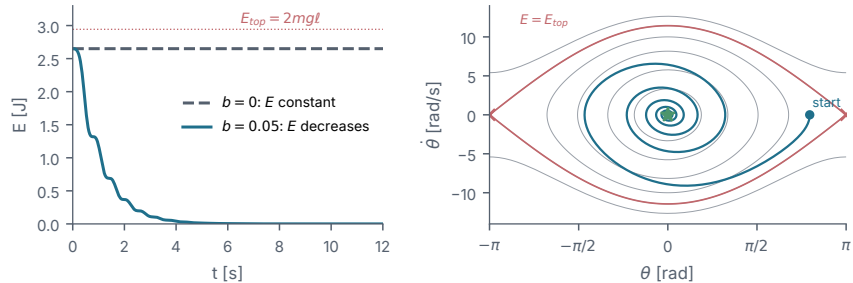
Step 3. The gravity terms cancel:

$$\boxed{\dot E = -\,b\,\dot\theta^2 + u\,\dot\theta}$$

- $-b\dot\theta^2 \le 0$: friction can only **remove** energy;
- $u\dot\theta$ is the **power** of the motor: it adds energy when it pushes in the direction of motion, and removes it when it pushes against.

In `pendulum_lyapunov.py`, mark `[ENERGY]`: function `energy`. Run `'free'` ($u = 0$, released at 2.5 rad) and `'free_no_fric'` (same, with $b = 0$). The middle plot is $E(t)$.

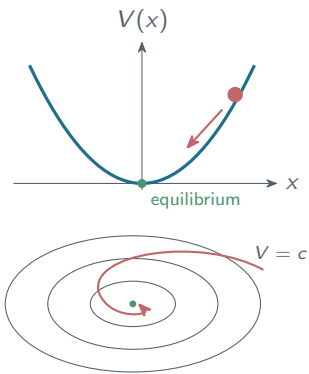
What the simulation shows



- **Left.** With friction, E only goes down, in steps: flat at the turning points ($\dot{\theta} = 0$), steepest at the bottom. Without friction, $E = 2.650$ J forever.
- **Right.** Curves of constant E in the plane $(\theta, \dot{\theta})$. Since friction dissipates energy, the trajectory moves to lower and lower levels, and ends at the bottom (star). It can never reach the red curve $E = E_{top}$, which passes through the top (crosses).

Two facts, no solution of the equation: E is a bowl with its bottom at the rest point, and E never goes up. That already says the pendulum stays near the bottom (Lyapunov's idea). To prove that it actually *stops* there, we will need one more argument (LaSalle).

Lyapunov's idea



To study an equilibrium, do not solve the equations. Find a function $V(x)$ that is a **bowl**:

- $V = 0$ at the equilibrium, $V > 0$ everywhere else nearby;

and check that the motion only goes **downhill**:

- $\dot{V} \leq 0$ along every solution.

Then the state cannot climb out: once inside a level curve $V = c$, it stays inside (**stability**). Only if $\dot{V} < 0$ except at the equilibrium is it sure to go down to the bottom (**convergence**).
Such a V is a **Lyapunov function**. For mechanical systems, try the energy first.

Lyapunov's theorem, and how to compute $\dot{V}$

System $\dot{x} = f(x)$ with an equilibrium at $x = 0$ (move it there with $e = x - x^*$). Take $V(x)$ with $V(0) = 0$ and $V(x) > 0$ for $x \neq 0$ in a region around 0.

If, in that region,	then the equilibrium is
$\dot{V} \leq 0$	**stable**: starting close, the state stays close
$\dot{V} < 0$ for $x \neq 0$	**asymptotically stable**: it also goes to 0
$V > 0$ and $\dot{V} < 0$ for all $x \neq 0$, and $V \to \infty$ as $\lvert x \rvert \to \infty$	**globally** asymptotically stable

Computing $\dot{V}$. Chain rule, then replace $\dot{x}$ by $f(x)$, as we did for E:

$$\dot{V} = \frac{\partial V}{\partial x_1}\dot{x}_1 + \cdots + \frac{\partial V}{\partial x_n}\dot{x}_n = \nabla V(x) \cdot f(x).$$

No solution of the equation is needed: only f, which we always have. **Warning.** If your candidate V fails, that proves nothing: the system may still be stable, and a better V may exist. Failing with one candidate never proves instability.

Exercise 1: where linearization says nothing

Use $V = \frac{1}{2}x^2$ for the two scalar systems below.

1. $\dot{x} = -x^3$. Linearize at $x = 0$: what is the eigenvalue, and what does it tell you? Now compute $\dot{V}$. Conclusion? Does x go to 0 exponentially?
2. $\dot{x} = -x + x^3$. Linearization says $x = 0$ is stable. But from how far? Compute $\dot{V}$ and find the region where $\dot{V} < 0$. What happens if $x(0) = 2$?
3. In both cases you guessed $V = \frac{1}{2}x^2$. If it had failed, would that prove the equilibrium unstable?

Hint. $\dot{V} = x\,\dot{x}$. For a scalar system the linearization is $\dot{\delta x} = f'(0)\,\delta x$ (Session 3).

Exercise 1: solution

Step 1. $f(x) = -x^3$, $f'(0) = -3x^2|_0 = 0$. Eigenvalue zero: the linear model $\dot{\delta x} = 0$ says "nothing moves", which is wrong. Linearization is **inconclusive**. With V: $\dot{V} = x(-x^3) = -x^4 < 0$ for $x \neq 0$, everywhere, and $V \to \infty$. Globally asymptotically stable, but not exponentially: solving, $x(t) = x_0/\sqrt{1 + 2x_0^2 t}$, which decays like $1/\sqrt{2t}$, slower than any $e^{-\lambda t}$.

Step 2. $f'(0) = -1 + 3x^2|_0 = -1$: stable near 0, and linearization says no more.
$\dot{V} = x(-x + x^3) = -x^2(1 - x^2)$, negative for $0 < |x| < 1$. Any start with $|x(0)| < 1$ goes to 0. From $x(0) = 2$: $\dot{x} = -2 + 8 = 6 > 0$, x grows and escapes (the solution blows up in finite time). Indeed $x = \pm 1$ are two other equilibria: the region cannot be larger.

Step 3. No. A failed V only means a bad guess; another V might work. To prove instability you need a different argument (for example, an eigenvalue with positive real part).

Lesson. Lyapunov answers what linearization cannot: cases with a zero eigenvalue, and an estimate of *how large* the region of attraction is (in this scalar case, the exact one).

The trouble with $\dot{V} \leq 0$

For the free pendulum ($u = 0$), take $V = E$. Then

$$\dot{V} = -b\,\dot{\theta}^2 \leq 0.$$

It is **not** < 0 for every state $\neq 0$: $\dot{V} = 0$ at every point with $\dot{\theta} = 0$, for example the turning points of each swing, where $\theta \neq 0$.

So the theorem only gives *stable*: the pendulum stays near the bottom. It does not say that the pendulum stops. Yet the simulation clearly shows it does.

The way out. The pendulum can be at $\dot{\theta} = 0$ only for an instant: at a turning point gravity immediately makes it move again, and E drops again. The only place where it can *stay* with $\dot{\theta} = 0$ is an equilibrium. That is LaSalle's argument.

The same problem appeared in Session 5: $\dot{V} = -k_d \dot{e}^2$ is zero whenever $\dot{e} = 0$.

LaSalle's argument

LaSalle (informal)

Suppose the motion stays in a compact set Ω it cannot leave (positively invariant, e.g. a bounded level set $V \le c$), and $\dot{V} \le 0$ in Ω. Then the state ends in the largest set of states where it can *stay forever* with $\dot{V} = 0$.

Recipe, for the free pendulum ($b > 0$; angle counted mod 2π; the speed is bounded because E never grows).

Step 1. Where is $\dot{V} = 0$? Where $\dot{\theta} = 0$.

Step 2. Stay there forever: $\dot{\theta}(t) \equiv 0$, so $\ddot{\theta} \equiv 0$.

Step 3. Put that in the model: $0 = -mg\ell \sin\theta - 0$, so $\sin\theta = 0$: $\theta = 0$ or $\theta = \pi \pmod{2\pi}$.

Step 4. It ends at the bottom or exactly at the top. Released from 2.5 rad, $E(0) = 2.650\,\text{J} < E_{top} = 2.943\,\text{J}$: it cannot be at the top (E would have to rise), so it ends at the bottom. (Exactly upright at rest, it stays: `'free_top'`.)

LaSalle answers Session 5's question

Computed torque with a perfect model: $\ddot{e} + k_d\dot{e} + k_p e = 0$, and

$V = \frac{1}{2}\dot{e}^2 + \frac{1}{2}k_p e^2$ (a bowl, zero only at $e = \dot{e} = 0$), $\qquad \dot{V} = \dot{e}\,\ddot{e} + k_p e\,\dot{e} = \dot{e}(-k_d\dot{e} - k_p e) + k_p e\,\dot{e} = -k_d\dot{e}^2 \leq 0.$

Step 1. $\dot{V} = 0$ where $\dot{e} = 0$.

Step 2. Stay there: $\dot{e} \equiv 0$, so $\ddot{e} \equiv 0$.

Step 3. The error equation gives $0 + 0 + k_p e = 0$, so $e = 0$.

Step 4. The only state where it can stay is $e = \dot{e} = 0$: the error goes to zero, from every initial condition. □

Not one simulation was needed, and nothing was linearized.

The pattern. $\dot{V} \leq 0$ says "the system does not climb". LaSalle adds "and it cannot rest anywhere except at the equilibrium". Together: it goes to the equilibrium.

Exercise 2: same $\dot{V}$, different ending

Two systems, the same candidate $V = \frac{1}{2}(x_1^2 + x_2^2)$.

$$\text{A: } \dot{x}_1 = x_2, \quad \dot{x}_2 = -x_1 - x_2 \qquad\qquad \text{B: } \dot{x}_1 = 0, \quad \dot{x}_2 = -x_2$$

1. Compute $\dot{V}$ for both. What can Lyapunov's theorem (the table) conclude?
2. Apply LaSalle's recipe to each. Which one goes to the origin from every start? Where does the other one end?
3. Remove the damping in A: $\dot{x}_2 = -x_1$. What does $\dot{V}$ tell you now, and what does the system do?

Hint. A is a mass, a spring and a damper. In step 3 of the recipe, use the equations of *each* system.

Exercise 2: solution

Step 1. A: $\dot{V} = x_1 x_2 + x_2(-x_1 - x_2) = -x_2^2$. B: $\dot{V} = x_1 \cdot 0 + x_2(-x_2) = -x_2^2$. Identical, and only ≤ 0 (zero on the whole axis $x_2 = 0$): the theorem gives *stable* for both, nothing more.

Step 2. Stay forever with $x_2 \equiv 0$, so $\dot{x}_2 \equiv 0$. A: $0 = -x_1 - 0$, so $x_1 = 0$: only the origin. Every start goes to 0 (V is a bowl that grows to infinity, so the motion stays bounded). B: $\dot{x}_2 = 0$ holds for *any* x_1, and $\dot{x}_1 = 0$ keeps x_1 where it started. Every point $(c, 0)$ is an equilibrium: B ends at $(x_1(0), 0)$, not at the origin.

Step 3. $\dot{V} = x_1 x_2 - x_2 x_1 \equiv 0$. The state moves on a circle $V = \text{const}$ forever: stable, not asymptotically stable. LaSalle cannot conclude convergence with this V, and indeed there is none (the free pendulum with $b = 0$ does the same).

Lesson. $\dot{V} \leq 0$ alone does not decide. What decides is step 3 of the recipe: what the dynamics allow when $\dot{V} = 0$.

From analysis to design

So far V was used to *check* a system. Now we use it to *build* the controller. Remember $\dot{E} = -b\dot{\theta}^2 + u\,\dot{\theta}$: the input u appears in $\dot{V}$. We can choose it to make $\dot{V} \leq 0$.

Lyapunov-based design, four steps

1. Choose V: zero at the goal, positive elsewhere (an energy is a good start).
2. Compute $\dot{V}$ with the model. The input u appears.
3. Choose u so that $\dot{V} \leq 0$, keeping the terms that already help.
4. Use LaSalle to check that the state can only rest at the goal.

Step 3 alone ($\dot{V} \leq 0$) only gives stability. Step 4, or $\dot{V} < 0$, is what gives convergence to the goal.

Module 1 chose eigenvalues, valid near one point. Here we choose a function whose decrease we verify at *every* state where we computed it. Two designs: hold any angle (1), swing up with a weak motor (2).

Design 1: hold any angle

Goal: hold θ_d (constant). Error $e = \theta - \theta_d$, so $\dot{e} = \dot{\theta}$, $\ddot{e} = \ddot{\theta}$. The error equation, with $\theta = e + \theta_d$:

$$m\ell^2\ddot{e} = -mg\ell\sin(e + \theta_d) - b\dot{e} + u.$$

Step 1. Choose $V = \frac{1}{2}m\ell^2\dot{e}^2 + \frac{1}{2}m\ell^2 k_p e^2$ (kinetic energy plus a "spring" on the error).

Step 2. $\dot{V} = \dot{e}\,(m\ell^2\ddot{e}) + m\ell^2 k_p e\dot{e} = \dot{e}\left[-mg\ell\sin(e + \theta_d) - b\dot{e} + u + m\ell^2 k_p e\right]$.

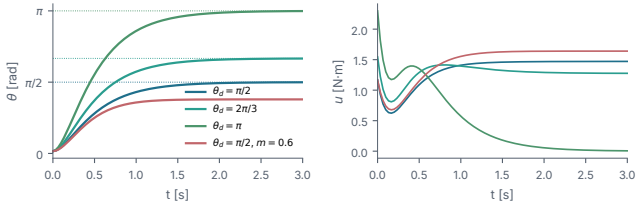
Step 3. Cancel gravity, cancel the spring term, add damping:

$$\boxed{u = mg\ell\sin(e + \theta_d) - m\ell^2\left(k_p e + k_d\dot{e}\right)} \quad \Longrightarrow \quad \dot{V} = -\left(b + m\ell^2 k_d\right)\dot{e}^2 \leq 0.$$

Step 4. LaSalle: $\dot{e} \equiv 0 \Rightarrow \ddot{e} \equiv 0 \Rightarrow m\ell^2 k_p e = 0 \Rightarrow e = 0$. The pendulum goes to θ_d from *every* start, under the conditions of this proof: exact model, no torque limit, $k_p > 0$, $b + m\ell^2 k_d > 0$, and e the continuous angle difference.

In `pendulum_lyapunov.py`, mark `[PDG]`: `pd_gravity` wraps e to $[-\pi, \pi)$ (shorter way). The global proof is for the continuous error; with wrap it holds locally, in regions that do not cross $e = \pm\pi$.

Design 1: what it does



- From hanging at rest, it holds $\pi/2$, $2\pi/3$ and the top. The final torque is the holding torque $mg\ell \sin\theta_d$: 1.47 N m at the horizontal, 0 at the top.
- **Friction was not cancelled**, unlike Session 5: $-b\dot{e}^2$ already helps $\dot{V}$. Lyapunov design keeps the terms that help and fights only those that hurt.
- **Wrong mass** (red, $m = 0.6$): it stops at 1.191, an offset of -0.38, as in Session 5. Step 3 cancelled gravity with $\hat{m} = 0.5$; the leftover term breaks $\dot{V} \leq 0$. How to deal with it: next session.

In `pendulum_lyapunov.py`, **mark** `[PDG]`: run `'hold_side'`, `'hold_120'`, `'hold_top'` (peak torque 2.3 N m: the 3 N m limit is not reached, so the unsaturated analysis applies along these runs). The middle plot is $V(t)$: it never goes up. Then `'hold_wrong_m'`: V stops at 0.052, not 0. Which step failed?

Exercise 3: cancel or dominate?

Scalar system $\dot{x} = a\,x^2 + u$. The model says $a = 1$. Use $V = \frac{1}{2}x^2$, $k > 0$.

1. Cancel: $u = -x^2 - kx$. Compute $\dot{V}$ when the model is right, $a = 1$.
2. The real system has $a = 1.5$. Same controller. Compute $\dot{V}$: where is it negative? What happens if $x(0)$ is large?
3. You only know $|a| \le 2$. Find a u that gives $\dot{V} \le -kx^2$ for *every* such a.

Hint for 3. You cannot cancel ax^2 without knowing a, but you can overpower it: add a term that grows like x^2, always opposes x, and is at least as large as $|a|x^2$.

Exercise 3: solution

Step 1. $\dot{x} = x^2 - x^2 - kx = -kx$, $\dot{V} = -kx^2$: globally asymptotically stable.

Step 2. $\dot{x} = 0.5x^2 - kx$, so $\dot{V} = x\dot{x} = x^2(0.5x - k)$. Negative only for $x < 2k$. Starting beyond $x = 2k$ (another equilibrium), x grows and escapes. A 50% model error turned a global result into a local one.

Step 3. Take $u = -kx - 2x|x|$. Then

$$\dot{V} = x\big(ax^2 - 2x|x| - kx\big) = a\,x^3 - 2|x|^3 - kx^2 \le |a|\,|x|^3 - 2|x|^3 - kx^2 \le -kx^2,$$

for every $|a| \le 2$. Global, with no knowledge of a beyond its bound.

Lesson. Cancelling a nonlinearity requires knowing it exactly. Dominating it only requires a bound, and V tells you how much is enough. This is the idea behind robust control (Session 7).

Design 2: the problem

Goal. Bring the pendulum from hanging at rest to upright at rest, with a weak motor: $|u| \le u_{\max} = 1.2\,\mathrm{N\,m}$.

Why it is hard. Holding the pendulum horizontal needs $mg\ell = 1.47\,\mathrm{N\,m}$, more than the motor gives. Pushing it straight up is impossible (Session 5 tried with a path: it failed).

Idea. Swing it up, as you push a child on a swing: push *in the direction of motion*, a little each swing, until the pendulum has the energy of the upright position. Then catch it.

The equations we need

$$E = \tfrac{1}{2} m\ell^2 \dot\theta^2 + mg\ell(1 - \cos\theta), \qquad E_{top} = 2mg\ell = 2.943\,\mathrm{J} \ \text{(upright, at rest)},$$

$$\dot E = u\,\dot\theta - b\,\dot\theta^2 \qquad \text{(the chain rule slide)}.$$

So the motor adds energy when u has the sign of $\dot\theta$. The task becomes: bring E to E_{top}.

Design 2: the controller, from V

Step 1. Choose $V = \frac{1}{2}\left(E - E_{top}\right)^2$: zero when the energy is right.

Step 2. $\dot{V} = \left(E - E_{top}\right)\dot{E} = \left(E - E_{top}\right)\left(u\,\dot{\theta} - b\,\dot{\theta}^2\right)$.

Step 3. Friction now *hurts*: below the top energy ($E < E_{top}$) the term $-\left(E - E_{top}\right) b\dot{\theta}^2$ is positive. So cancel it, and add the pump:

$$u = \left(b - k\left(E - E_{top}\right)\right)\dot{\theta} \quad \Longrightarrow \quad \dot{V} = -k\left(E - E_{top}\right)^2 \dot{\theta}^2 \leq 0, \qquad k = 20.$$

(In Design 1 friction helped, so we kept it: keep what helps, cancel what hurts.)

Reading the law. Below the top energy, u has the sign of $\dot{\theta}$: push with the motion. $\dot{V} = 0$ only at the turning points ($\dot{\theta} = 0$) or when $E = E_{top}$.

Motor limit. The motor applies $\mathrm{sat}(u)$, at most 1.2 N m. The push keeps its sign and still beats friction: $u_{\max}|\dot{\theta}| > b\dot{\theta}^2$ for $|\dot{\theta}| < u_{\max}/b = 24$ rad/s, far above the speeds of the swing-up (at most 11.4 rad/s).

In `pendulum_lyapunov.py`, mark `[SWING]`: in `controller`, the last two lines.

Design 2: catching at the top

$V = 0$ on the whole curve $E = E_{top}$, not only at the top. Reaching the right energy, the pendulum creeps toward the top slower and slower (it never gets there in finite time), and the top is unstable: any disturbance throws it off. We need a second controller that holds the angle.

Switching rule. Pump until the pendulum is near the top and slow,

$$|\mathrm{wrap}(\theta - \pi)| < 0.3\,\mathrm{rad}\ (17^\circ) \quad \text{and} \quad |\dot{\theta}| < 3\,\mathrm{rad/s},$$

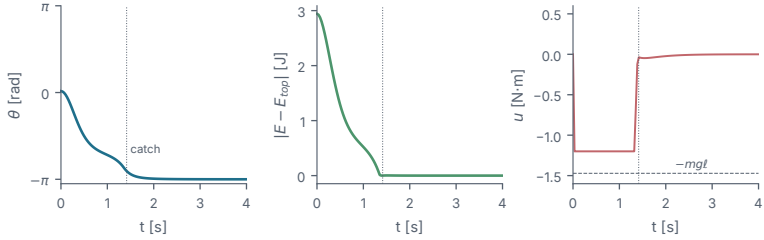
then switch to Design 1 with $\theta_d = \pi$ (limited to $u_{\max}$), and never switch back. (wrap maps angles to $[-\pi, \pi)$; $-\pi$ and π are the same upright position.)

Why Design 1 can hold it with a weak motor

- near the top, gravity's torque $mg\ell \sin\theta$ is small;
- with $E \approx E_{top}$, the pendulum arrives almost at rest.

In `pendulum_lyapunov.py`, mark `[CATCH]`: in `simulate`, the lines with `near`.

Design 2: what happens



- The motor pushes at its limit, $-1.2\,\mathrm{N\,m}$, in the direction the pendulum is already falling; momentum carries it past the horizontal. $|E - E_{top}|$ only goes down, and the pendulum is caught at 1.42 s, then held upright with $u \approx 0$.
- **Same motor as Session 5, different question**: there, follow a path against gravity; here, reach an energy and let gravity help.

In `pendulum_lyapunov.py`, mark `[SWING]`: run `'swing_up'`, then `'swing_weaker'` (0.6 N m, caught at 2.75 s).

Exercise 4: when the swing-up fails

Run each scenario, then explain with $u = \mathrm{sat}\big((b - k(E - E_{top}))\dot{\theta}\big)$ and $\dot{E} = u\dot{\theta} - b\dot{\theta}^2$.

1. `'swing_at_rest'`: the pendulum starts *exactly* hanging at rest. The motor never moves. Why? How would you fix it?
2. `'swing_too_weak'`: $u_{\max} = 0.35\,\mathrm{N\,m}$. The energy stops growing about 1 J short of E_{top}. Which term of $\dot{E}$ wins, and where in the swing? Would a larger k help?

Hint for 2. At the bottom, how fast must the pendulum move to have $E = E_{top}$?

Exercise 4: solution

Step 1. $\dot{\theta} = 0$ gives $u = 0$, and at the bottom gravity gives no torque either: it is an equilibrium, nothing moves. Fix: a small initial push (the default start is $\theta_0 = 0.05\,\text{rad}$).

Step 2. At the bottom, $\frac{1}{2} m\ell^2 \dot{\theta}^2 = E_{top}$ gives $\dot{\theta} = 2\sqrt{g/\ell} = 11.4\,\text{rad/s}$. The motor adds at most $u_{\max}|\dot{\theta}|$; friction removes $b\dot{\theta}^2$. Friction wins when $|\dot{\theta}| > u_{\max}/b = 7\,\text{rad/s}$: near the bottom, each swing loses what it gained. A larger k does not help, the motor is already at its limit. Fix: a stronger motor (0.4 N m just works) or less friction.

Lesson. $\dot{E}$ explains every failure in one line, and points to the fix.

Module 1 versus today

	Linearization + eigenvalues	Lyapunov
Answers	near one equilibrium	in a region you can estimate, sometimes everywhere
You need	Jacobians and eigenvalues	a function V (try the energy first)
Design	place eigenvalues of $A - BK$	choose V, then u with $\dot{V} \leq 0$
Saturation	outside the theory	included, if $\dot{V} \leq 0$ still holds with $\lvert u \rvert \leq u_{\max}$
Model terms	all linearized	keep those that help (friction)
Weak point	only local	finding V is an art; no V found proves nothing

Both are used together in practice: energy control to bring the system into a region suitable for local stabilization (swing-up), a linear or simple controller to hold it there (the catch). And near an equilibrium whose linearization is Hurwitz (all eigenvalues with negative real part), a quadratic Lyapunov function always exists, so the two agree where both apply.

Lab 6: running your own scenario

Same procedure as Labs 4 and 5. At the end of `SCENARIOS` in `pendulum_lyapunov.py`, add one line per case, for example

```
    'A1_free2':   dict(law='none', theta0=2.0),
    'C5_u05':     dict(law='swing', umax=0.5),
```

set `SCENARIO` to that name, and run. Each run prints the final angle, the final V, the largest increase of V, the peak torque and, for the swing-up, the catch time. It saves the plots to `<name>.png` and shows the animation (15 s).

Key	Meaning	Default	Where on the slides
`law`	`'none'` free, `'pdg'` Design 1, `'swing'` Design 2	`'swing'`	
`theta0`	initial angle, at rest [rad]	0.05	energy, Exercise 4
`b`	friction	0.05	energy, LaSalle
`target`	angle held by `'pdg'` [rad]	π	Design 1
`kp`, `kd`	gains of Design 1 (and of the catch)	16, 8	Design 1
`m`	real mass [kg]; the controller believes 0.5	0.5	Design 1
`k`	pumping gain	20	Design 2
`umax`	motor limit for `'swing'` [N·m]	1.2	Design 2
`catch`	switch to Design 1 near the top	`True`	the catch

Lab 6: what to do

In every item, **predict first** (one sentence, using $\dot{V}$), then run, then explain any difference between prediction and result.

Part A: energy

1. Run `law='none'` with `theta0` $= 1,\ 2,\ 2.5,\ 3$, and with `b=0`. Table: $E(0)$ by hand, next to the first value of the V plot. Is there any $\theta_0 < \pi$, at rest, from which the free pendulum passes over the top? Explain with $\dot{E}$.

Part B: Design 1

2. Hold $\pi/2$ with `kd=0`; then with `kd=0` and `b=0`. Does it settle? Explain both with $\dot{V} = -(b + m\ell^2 k_d)\dot{e}^2$ and LaSalle.
3. Hold $\pi/2$ with `m` $= 0.45,\ 0.55,\ 0.6$. Does V go to zero? Which step of the Design 1 proof no longer holds, and why?

Part C: Design 2

4. Swing up with `umax` $= 1.2,\ 0.8,\ 0.6,\ 0.5,\ 0.4,\ 0.35$. Table: catch time. Which motors work within the 15 s of the simulation? Compare with $mg\ell$, and explain the failures with Exercise 4.

Lab 6 is Homework 6, items 1 to 6. No new code is needed.

Lab 6: your own design

Part D (paper only, no simulation)

5. For $\dot{x}_1 = x_2,\ \dot{x}_2 = -\sin x_1 + u$, choose a V, design u, and prove that the origin is asymptotically stable (LaSalle). Then the motor is limited, $|u| \le u_{\max}$: does your proof still hold? Say exactly which step fails, and where in the plane. Can you find a region, a level set $V \le c$, where your u never saturates, so the proof still holds there?
6. Final question: list the claims of Parts A to D and classify each as *proven* (Lyapunov, LaSalle), *observed* (simulation only) or *not established*.

Hint for 5. Design 1 is a good model: the pendulum with $m\ell^2 = 1$, $g/\ell = 1$, $b = 0$.

Every answer is a table, a printed number, an image saved by the program, a short derivation, or a few sentences.

Wrap-up and next session

Today

- the pendulum's energy obeys $\dot{E} = -b\hat{\theta}^2 + u\hat{\theta}$: friction removes, the motor adds power $u\hat{\theta}$;
- Lyapunov: a bowl V with $\dot{V} \leq 0$ shows stability without solving the equations, and helps estimate a region of attraction;
- LaSalle: with $\dot{V} \leq 0$, the state ends where it can stay with $\dot{V} = 0$; this proved $e \to 0$ for Session 5;
- design: choose V, then u. PD + gravity compensation holds any angle; energy pumping regulates E, and with a catch swings up a motor weaker than $mg\ell$.

Next session: robustness and observers

- the wrong mass again: dominate instead of cancel (Exercise 3), and how large an error a controller tolerates;
- estimating what we do not measure: velocity, and an unknown disturbance;
- controllers built on estimates.

Homework 6: due before the next session

Complete items 1 to 6 of Lab 6 and submit a **single PDF** containing the requested tables, images and brief answers. Upload the PDF to the *Homework 6* assignment on Blackboard by the posted deadline. Submissions by email will not be accepted.